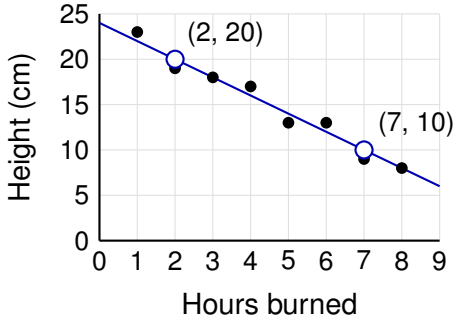
Unit 4 Review KEY

Name _____

One data set, every tool. 8 candles of the same kind burned for different numbers of hours. Then each candle was measured.

Hours burned (x)	1	2	3	4	5	6	7	8
Height in cm (y)	23	19	18	17	13	13	9	8

4.1 Describe and fit.
Direction, form, strength.
A line of best fit goes through the middle of the dots.



Negative, linear, strong.

Two points on the line: $(2, 20)$ and $(7, 10)$.

$$m = \frac{10 - 20}{7 - 2} = \frac{-10}{5} = -2$$

$20 = -2(2) + b$, so $b = 24$

$\hat{y} = -2x + 24$

Predict for 4.5 hours: $\hat{y} = -2(4.5) + 24 = 15$ cm

Predict for 15 hours: $\hat{y} = -2(15) + 24 = -6$ cm. A height cannot be negative. The data only goes from 1 to 8 hours. This is **extrapolation**.

4.2 Explain the line.
Slope: for each additional [x unit], the predicted [y] changes by the slope.
r is from -1 to 1.

Slope (-2): For each additional hour, the predicted height decreases by 2 cm.

y-intercept (24): When $x = 0$ (a new candle), the predicted height is 24 cm.

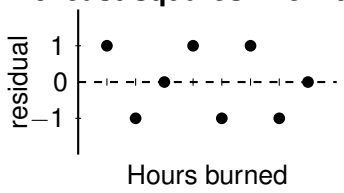
r: The dots are close to a line that goes down. r is close to -1 ($r \approx -0.98$).

4.3 Find a residual.
residual
= actual $-$ predicted
$= y - \hat{y}$
Positive: above the line.
Negative: below the line.

The candle that burned 4 hours is 17 cm. $\hat{y} = -2(4) + 24 = 16$ cm

Residual $= 17 - 16 = 1$. It is positive, so the dot is **above** the line. The candle is 1 cm taller than predicted.

The **least-squares line** makes $\Sigma(y - \hat{y})^2$ as small as possible.



The residual plot shows no pattern. A line is a **good** model.

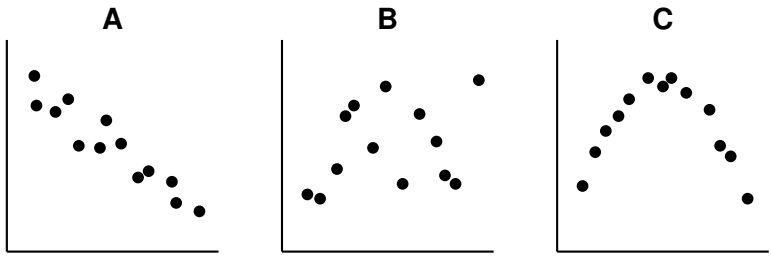
Correlation is not causation. When more umbrellas are sold, there are more car accidents. Umbrellas do not cause accidents. Rain causes both. Rain is a **lurking variable**.

Work with a partner. Compare answers when you finish each problem.

4.1 - Scatterplots and Lines of Best Fit

1. **I describe the direction, form, and strength of a scatterplot, and I choose a good line of best fit.**

   Describe the direction, form, and strength of each scatterplot.



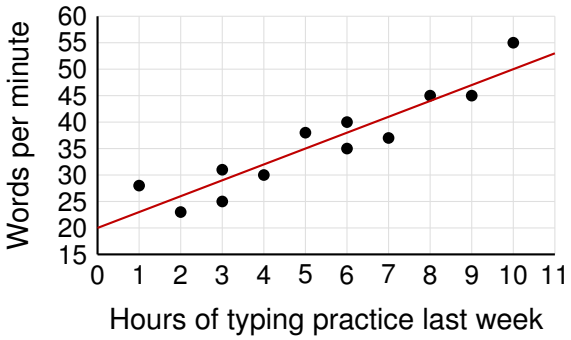
   (a) A Negative, linear, strong

   (b) B Positive, linear, weak (the dots are spread far from a line)

   (c) C Curved (not linear), strong. It goes up and then down.

   12 students recorded how many hours they practiced typing last week and their typing speed.

Hours (x)	1	2	3	3	4	5	6	6	7	8	9	10
Words per minute (y)	28	23	31	25	30	38	35	40	37	45	45	55



   (d) Describe the direction, form, and strength.

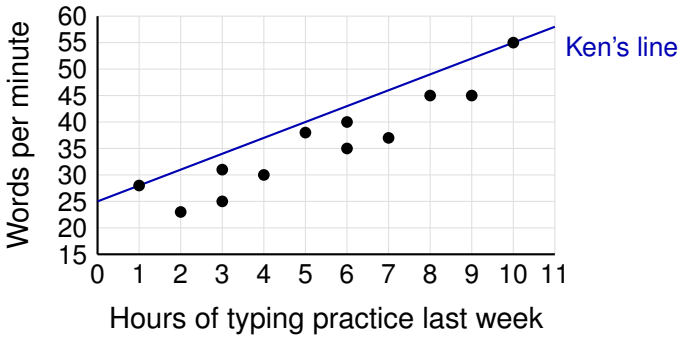
   Positive, linear, strong

   (e) Draw a line of best fit on the graph.

   One good line is shown in red. Accept any line that follows the trend with about as many dots above as below.

   (f) How many dots are above your line? How many are below?

   Red line: 6 above and 6 below. About the same number is the goal.

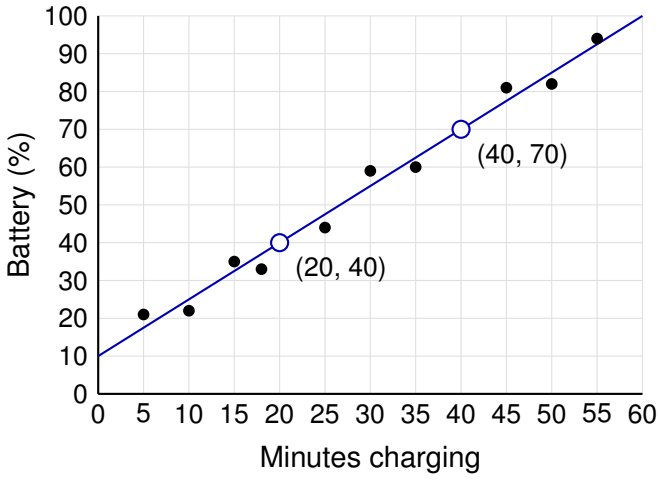


   (g) Ken drew this line for the same data. It goes through the first dot and the last dot. Is it a good line of best fit? Explain.

   No. 10 dots are below it and none are above it. A good line has about the same number of dots above and below.

2. **I write the equation of a line of best fit, and I make predictions and know when not to trust them.**

Mia charged her phone on 10 days. She recorded how many minutes it charged and the battery percent at the end. A line of best fit is drawn.



(a) Describe the direction, form, and strength.

Positive, linear, strong

(b) Use the two marked points to find the slope.

$m = \dfrac{70 - 40}{40 - 20} = \dfrac{30}{20} = 1.5$

(c) Find b.

$40 = 1.5(20) + b$

$40 = 30 + b$, so $b = 10$

(d) Write the equation of the line. $\hat{y} = 1.5x + 10$

(e) Predict the battery percent after 26 minutes of charging.

$\hat{y} = 1.5(26) + 10 = 39 + 10 = 49$, about 49%

(f) Predict the battery percent after 80 minutes of charging. Should you trust this prediction? Explain.

$\hat{y} = 1.5(80) + 10 = 130$. No. A battery cannot be more than 100%. 80 minutes is far outside the data (5 to 55 minutes).

Two points on another line of best fit are $(2, 40)$ and $(8, 10)$.

(g) Find the slope.

$m = \dfrac{10 - 40}{8 - 2} = \dfrac{-30}{6} = -5$

(h) Find b and write the equation of the line.

$40 = -5(2) + b$, $40 = -10 + b$, so $b = 50$. $\hat{y} = -5x + 50$

4.2 - Slope, Intercept, and Correlation

3. **I explain the slope of a line of best fit in context, and I explain the y-intercept in context, or tell why it has no meaning.**

   Students on a long hike started with full water bottles. The line $\hat{y} = -6x + 32$ predicts the water left, where x is the number of hours of hiking and y is the ounces of water left.

   (a) Explain the slope in context.
   For each additional hour of hiking, the predicted water left decreases by 6 ounces.

   (b) Explain the y-intercept in context.
   When $x = 0$ (at the start of the hike), the predicted water left is 32 ounces.

   A store sells TVs. The line $\hat{y} = 15x - 200$ predicts the price, where x is the screen size in inches and y is the price in dollars.

   (c) Explain the slope in context.
   For each additional inch of screen size, the predicted price increases by $15.

   (d) Does the y-intercept have a real meaning here? Explain.
   No. It says a TV with a 0-inch screen costs $-$200. No TV has a 0-inch screen, and a price cannot be negative.

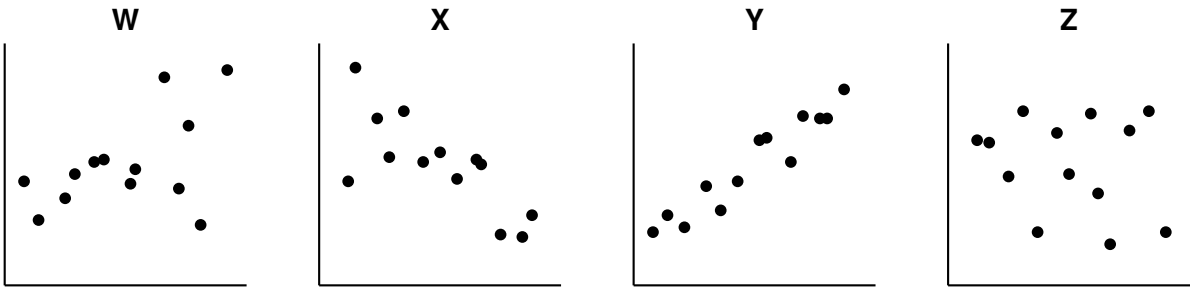
   A pizza shop uses the line $\hat{y} = 1.5x + 9$, where x is the number of toppings and y is the price of a pizza in dollars.

   (e) Explain the slope in context.
   For each additional topping, the predicted price increases by $1.50.

   (f) Explain the y-intercept in context.
   When there are 0 toppings (a plain cheese pizza), the predicted price is $9.

4. **I estimate the correlation coefficient r from a scatterplot, and I explain why correlation does not prove causation.**

(a) Match each r to a scatterplot: $r = 0.95$, $r = 0.5$, $r = -0.2$, $r = -0.75$



W: $r =$ 0.5 X: $r =$ −0.75 Y: $r =$ 0.95 Z: $r =$ −0.2

(b) Which shows a stronger linear relationship, $r = -0.7$ or $r = 0.6$? Explain.

$r = -0.7$. It is closer to -1 than 0.6 is to 1. The minus sign only shows the direction.

(c) Can a scatterplot have $r = 1.5$? Explain.

No. r is always from -1 to 1.

In an elementary school, students with bigger shoe sizes have higher reading scores. The correlation is strong and positive.

(d) Do bigger feet cause better reading? Explain.

No. A correlation shows that two things change together. It does not prove that one causes the other.

(e) What lurking variable could cause both?

Age (or grade). Older students have bigger feet, and they also read better.

In a city, the number of fans sold each day and the number of people at the pool have a strong positive correlation.

(f) Name a lurking variable that could cause both. Explain.

Hot weather (the temperature). On hot days, more people buy fans and more people go to the pool.

4.3 - Residuals

5. **I find and explain a residual, and I tell if a point is above or below the line from its residual.**

A delivery app predicts the delivery time with $\hat{y} = 4x + 10$, where x is the distance in miles and y is the time in minutes. Fill in the table.

x (miles)	y (actual minutes)	$\hat{y}$ (predicted)	residual $y - \hat{y}$
1	16	$4(1) + 10 = 14$	$16 - 14 = 2$
2	17	18	$17 - 18 = -1$
3	22	22	$22 - 22 = 0$
5	33	30	$33 - 30 = 3$
6	31	34	$31 - 34 = -3$
8	44	42	$44 - 42 = 2$

(a) Which orders have points above the line? How do you know?
The 1-mile, 5-mile, and 8-mile orders. Their residuals are positive.

(b) Which order has a point on the line? How do you know?
The 3-mile order. Its residual is 0.

(c) Explain the residual for the 6-mile order in context.
The residual is -3. The order took 3 minutes less than predicted.

(d) A 4-mile order has a residual of -2. How long did it take?
$\hat{y} = 4(4) + 10 = 26$. $y - 26 = -2$, so $y = 24$ minutes.

(e) Another order has a residual of 5. Is its point above or below the line? What does the residual mean?
Above, because the residual is positive. The order took 5 minutes longer than predicted.

6. **I explain what the least-squares line is, and I use a residual plot to decide if a line is a good model.**

Two lines were fit to the same 5 points. Here are their residuals.

	Residuals					$\Sigma(y-\hat{y})^2$
Line A	3	−2	1	−1	−1	$9+4+1+1+1=16$
Line B	1	−1	2	−2	0	$1+1+4+4+0=10$

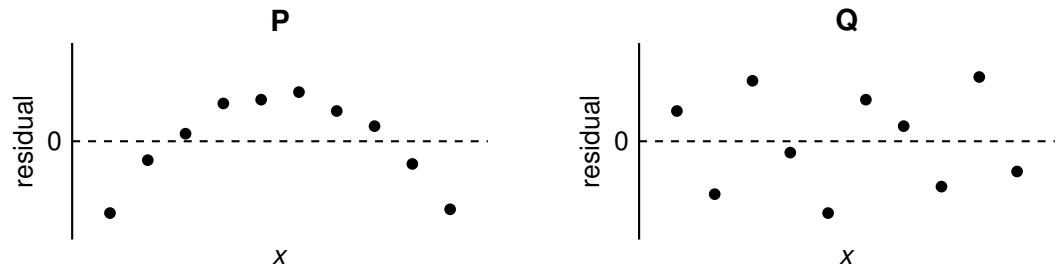
(a) Find $\Sigma(y-\hat{y})^2$ for each line. (Square each residual, then add.)

(b) Which line is closer to the least-squares line? Explain.
Line B. Its sum of squared residuals (10) is smaller than Line A's (16).

(c) Add the residuals of each line without squaring them. Why does this not help you choose a line?
Line A: $3-2+1-1-1=0$. Line B: $1-1+2-2+0=0$. Both sums are 0. The positive and negative residuals cancel out.

(d) For one data set, the least-squares line has $\Sigma(y-\hat{y})^2=12$. Can another line for the same data have $\Sigma(y-\hat{y})^2=9$? Explain.
No. The least-squares line has the smallest possible sum. Every other line has a sum of 12 or more.

These are the residual plots for two lines of best fit.



(e) Which plot shows that a line is a good model? Explain.
Plot Q. The residuals are scattered at random, with no pattern.

(f) What does the pattern in plot P tell you?
The residuals make a curve: negative, then positive, then negative. The data is curved, so a line is not a good model.

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

7. The scatterplot in Problem 1 shows the number of hours 12 students practiced typing last week and their typing speeds, in words per minute. Which equation best represents a line of best fit for the data?

   A $\hat{y} = -3x + 20$

   (B) $\hat{y} = 3x + 20$

   C $\hat{y} = 20x + 3$

   D $\hat{y} = 3x + 40$

8. Mia used the line of best fit $\hat{y} = 1.5x + 10$ to predict the battery percent, y, of her phone after it charges for x minutes. Her data included charging times from 5 to 55 minutes. Which prediction is **least** likely to be reliable?

   A the battery percent after 10 minutes of charging

   B the battery percent after 30 minutes of charging

   C the battery percent after 50 minutes of charging

   (D) the battery percent after 75 minutes of charging

9. A pet store owner found that the equation $\hat{y} = 2.5x + 20$ can be used to predict the price y, in dollars, of a fish tank that holds x gallons of water. Which statement best describes the meaning of the slope of the line?

   A Every fish tank costs \$2.50.

   (B) For each additional gallon a tank holds, the predicted price increases by \$2.50.

   C For each additional dollar in price, a tank holds 2.5 more gallons.

   D A tank that holds 0 gallons is predicted to cost \$2.50.

10. A study of 200 students found a strong positive correlation between the number of hours students practice a musical instrument each week and their math grades. Which conclusion is best supported by the study?

    A Practicing a musical instrument causes students to earn higher math grades.

    B Earning higher math grades causes students to practice a musical instrument more.

    (C) Students who practice more tend to have higher math grades, but the study does not show that one causes the other.

    D There is no relationship between practicing a musical instrument and math grades.

11. A bus company uses the line of best fit $\hat{y} = 3x + 12$ to predict the travel time y, in minutes, for a trip of x miles. A 6-mile trip took 27 minutes. Which statement about the residual for this trip is true?

A The residual is 3, and the point is above the line.

B The residual is 3, and the point is below the line.

C The residual is -3, and the point is above the line.

(D) The residual is -3, and the point is below the line.

$\hat{y} = 3(6) + 12 = 30$. Residual $= 27 - 30 = -3$. Negative, so the point is below the line.

12. A student found a line of best fit for a data set and made a residual plot. The residuals are positive for small and large values of x and negative for values of x in the middle. Which conclusion is best supported by the residual plot?

A A line is a good model, because some residuals are positive and some are negative.

(B) A line is not a good model, because the residuals show a curved pattern.

C A line is a good model, because the residuals form a clear pattern.

D A line is not a good model, because the sum of the residuals is 0.